Information Retrieval Indexing

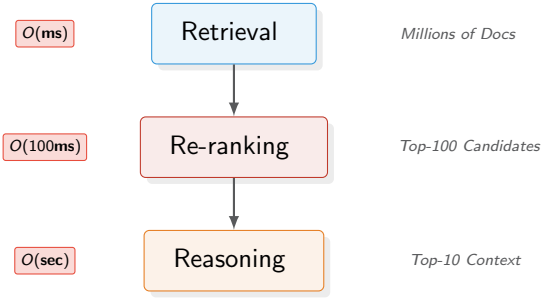
Lecture 2: Sparse and Dense Indexing

Avishek Anand

Information Retrieval

March 3, 2026

Retrieval → Re-ranking → Reasoning



Key Principle

Indexing sets the ceiling. Everything else (re-ranking, LLM reasoning) can only refine the results that retrieval finds.

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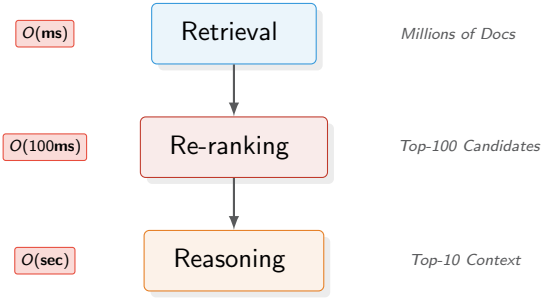

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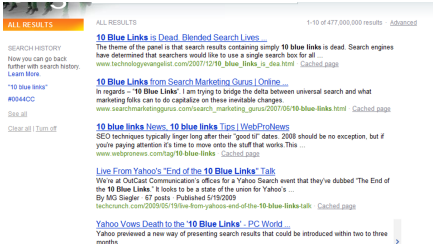

- **User Task:** Search for information.
- **Output:** A list of relevant documents.
- **Interaction:** User clicks links, reads, and synthesizes.
- **Classic System:** Google, Bing, Yahoo in the 2000s.

Problem

User does all the hard work of reading and combining information!

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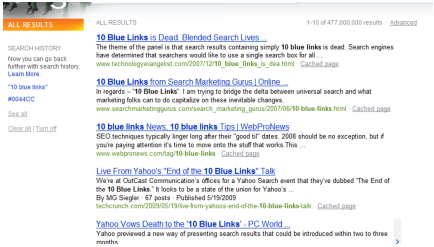

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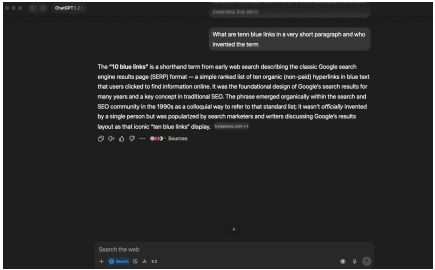
Modern IR: Retrieval-Augmented Generation (RAG)

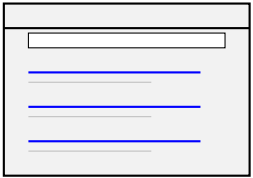


- **User Task:** Ask a question.
- **Output:** A synthesized answer in natural language.
- **Process:**
  1. **Retrieve** relevant facts.
  2. **Augment** LLM prompt with facts.
  3. **Generate** final answer.
- **Modern System:** ChatGPT, Perplexity, Bing Chat.

The Interface Changed. The Engine Did Not.

Old Interface (2000s):

- Output: Ranked documents
- Interaction: User reads/clicks



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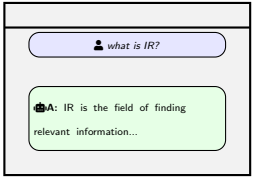

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Retrieval now serves models instead of users.
But the underlying components (Index, Retrieval, Relevance) remain the same.

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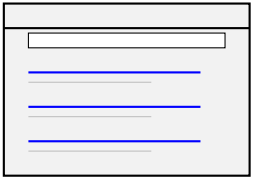

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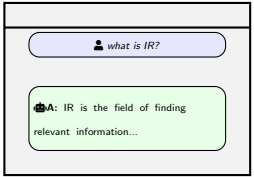

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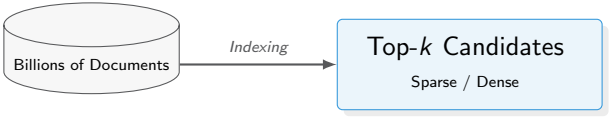
2. **Latency:** Users (and LLMs) expect sub-second responses.
3. **Feasibility:** Must reduce millions $\rightarrow$ hundreds of candidates.



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Indexing is not just an optimization. It is the bridge to feasibility.

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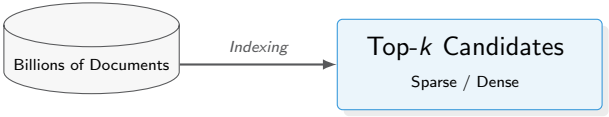

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Indexing Determines

- **Recall Ceiling:** Which documents can ever be retrieved.
- **Latency Budget:** How many candidates can be scored (fast vs slow).
- **Cost per Query:** Hardware requirements and compute cost.
- **What the LLM is allowed to see:** Retrieval is the LLM's "eyes".

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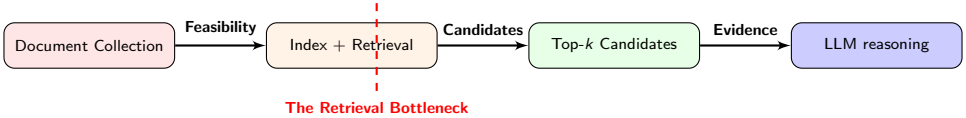

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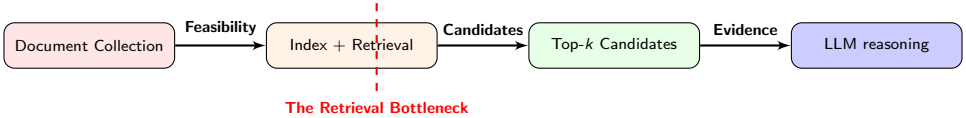

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Outline

1. Sparse Indexing: The Classical Approach
2. Dense Vector Indexing for Neural Retrieval
3. Design Decisions: When to Use What
4. Summary and Next Steps

The Inverted Index: Still Relevant in 2024

What Powers These Systems?

- Google Search
- Elasticsearch / OpenSearch
- Database full-text search (PostgreSQL, MySQL)
- RAG sparse retrieval component

Answer: **Inverted indexes**

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Answer: **Inverted indexes**

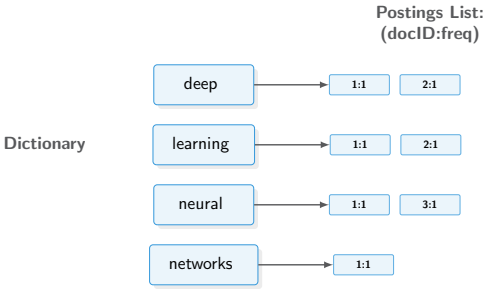
Why still relevant in the age of neural networks?

1. **Exact matching:** Find "iPhone 15 Pro Max" precisely
2. **Speed:** 5-10ms queries on millions of documents
3. **Explainability:** Can see which terms matched
4. **Hybrid search:** Complements dense retrieval

Inverted Index: Core Structure

Documents:

- Doc1: "neural networks for deep learning"
- Doc2: "deep learning applications"
- Doc3: "neural machine translation"



Query: "neural learning"
Process: Lookup "neural" → [1, 3], lookup "learning" → [1, 2]
Intersection: [1] ← Doc 1 contains both terms

Inverted Index as a Sparse Matrix

Vector Space Interpretation

- Documents = Rows (D)
- Terms = Columns (V)
- Values = Weights (TF-IDF, BM25)

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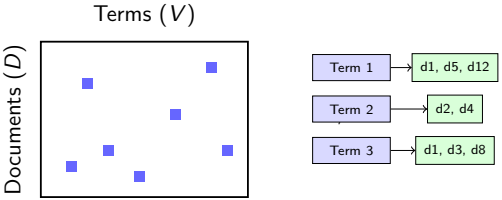

Key Insight

The Inverted Index is a **Compressed Sparse Column (CSC)** storage of the Document-Term matrix. We avoid storing zeros!

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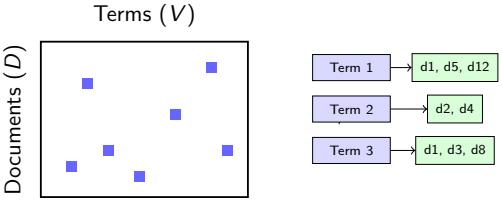

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Complexity: $O(N \cdot M)$

- N = number of documents
- M = avg document length

For 10M docs, 500 words each:
5 **billion** comparisons!

Inverted Index:

1. Lookup term in dictionary
2. Fetch posting list
3. Merge posting lists

Complexity: $O(k \cdot p)$

- k = query terms (typically 2-5)
- p = avg posting list size

For query "neural learning":
Lookup 2 terms, merge 1000 docs
$\rightarrow$ **1000$\times$ faster!**

Posting List Formats: Evolution

Level 1: DocID Only (Boolean Retrieval)

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neural:  [1, 3, 5, 12, 27, 89]
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- Supports: AND, OR, NOT queries
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- **This is what we use in practice**

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1. Tokenization

Simple: Split on whitespace
Reality: Language-dependent

- English: "don't" $\rightarrow$ "do" + "n't" or "dont"?
- German: "Informationssystem" $\rightarrow$ split compound?
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- running → run
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Stop Words: Modern Perspective

Traditional View (pre-2010):

- Remove common words: the, a, an, is, are, of, in, ...
- ✔ Smaller index
- ✔ Less noise

Problem:

- "to be or not to be" → ""
- "The Who" (band) → "Who"
- Loses phrase meaning

Modern View (2010+):

- **Keep stop words**
- Storage is cheap
- Important for phrases
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When to remove:

- Very large scale (web search)
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WordPiece Example (BERT):

```
[‘‘un’’, ‘‘##touch’’, ‘‘##able’’]
```

Breaks word into sub-tokens based on statistical frequency.

Index Compression: Why It Matters

The Bottleneck

Modern CPUs: 3+ GHz, billions of operations/second
Modern SSDs: 500 MB/s read speed

Decompression: 2-5 GB/s
Reading uncompressed: 0.5 GB/s

→ **Reading compressed data is faster!**

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Benefits:

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Original posting list (docIDs):

$$[5, 127, 214, 215, 512, 1024, 2048, 10239]$$

Observation: Most gaps between consecutive docIDs are small

Store gaps instead:

$$[5, 122, 87, 1, 297, 512, 1024, 8191]$$

Why This Helps

Original: Numbers range 5-10,239 $\rightarrow$ need 14 bits each
Gaps: Most numbers small (< 256) $\rightarrow$ need fewer bits

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Number	Binary	VB Encoding
5	101	1_0000101 (1 byte)
122	1111010	1_1111010 (1 byte)
824	1100111000	0_0000110 1_0111000 (2 bytes)

Example compression:

- Gaps: [5, 122, 87, 1] $\rightarrow$ 4 bytes (VB encoded)
- Uncompressed: 16 bytes (4 $\times$ 32-bit integers)
- **Compression ratio: 4$\times$**

VB Encoding: Step-by-Step Example

Encode 824:

1. Binary representation: 1100111000 (10 bits needed)
2. Needs 2 bytes ($10 > 7$)
3. Split into 7-bit chunks from right:
   - Lower 7 bits: 0111000
   - Upper bits: 0000110
4. First byte (higher bits): 0_0000110 (continuation = 0)
5. Second byte (lower bits): 1_0111000 (continuation = 1, last byte)
6. Result: 00000110 10111000

Decode:

1. Read bytes until continuation bit = 1
2. Strip continuation bits: 0000110 0111000
3. Concatenate: 0000110_0111000 = 1100111000 = 824 ✔

Compression in Practice

Typical Results

- VB encoding: 3-4$\times$ compression
- More sophisticated methods (PForDelta): 6-10$\times$ compression
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PyTerrier Demo: Sparse Indexing

```python
import pyterrier as pt
pt.init()

# Load dataset
dataset = pt.get_dataset('irds:vaswani')

# Index documents (builds inverted index with compression)
indexer = pt.IterDictIndexer("./vaswani_index", overwrite=True)
index_ref = indexer.index(dataset.get_corpus_iter())
print("Indexing complete!")

# Build retriever (uses BM25 scoring)
bm25 = pt.BatchRetrieve(index_ref, wmodel="BM25")

# Search
results = bm25.search("aerodynamic flow over wings")
print(results[['docno', 'score', 'rank']].head())

# Evaluate on full test set
evaluation = pt.Experiment(
    [bm25],
    dataset.get_topics(),
    dataset.get_qrels(),
    eval_metrics=["map", "ndcg_cut_10", "P_10"]
)
print(evaluation)
#    name       map  ndcg_cut_10      P_10
# 0  BM25  0.234567     0.456789  0.345678
```

This is production-ready sparse retrieval!

The Shift to Dense Vectors

Sparse (BM25)

Doc: "car problems"

Representation:

- Dimensions: 30,000 (vocab)
- car: 1
- problems: 1
- All else: 0
- **99.99% zeros**

Limitation:
Misses "automobile issues"

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Doc: "car problems"

Representation:

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- All non-zero
- **100% dense**

Advantage:
Matches "automobile issues"
(similar embedding!)

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Task

Given:

- Collection of N vectors: $\{\vec{d}_1, \vec{d}_2, \ldots, \vec{d}_N\}$
- Query vector: $\vec{q}$

Find: k vectors most similar to $\vec{q}$ (typically cosine similarity)

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Naive Solution: Exact Search

Compare query to **all** N vectors: $O(N \cdot d)$

For $N = 10^9$ docs, $d = 768$ dimensions:
768 billion operations $\rightarrow$ **too slow!** ($>$1 second even on GPU)

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Typical accuracy: 90-99% recall (good enough for IR!)

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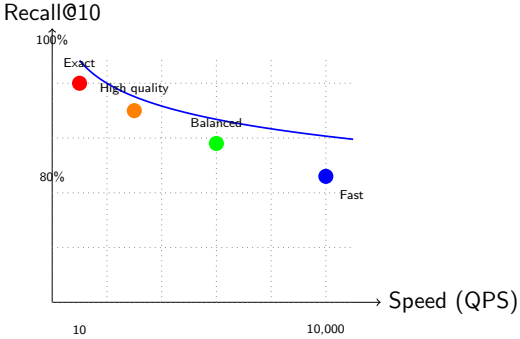
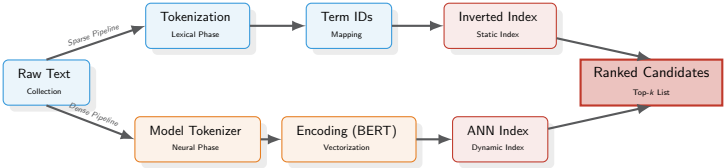
ANN: Accuracy-Speed Trade-off



Typical operating points:

- **High quality:** 95-99% recall @ 100-500 QPS (production RAG)
- **Balanced:** 90-95% recall @ 500-2000 QPS (most use cases)
- **Fast:** 80-90% recall @ 2000-10,000 QPS (first-stage retrieval)

From Text to Index: A Unified View



Key Distinction

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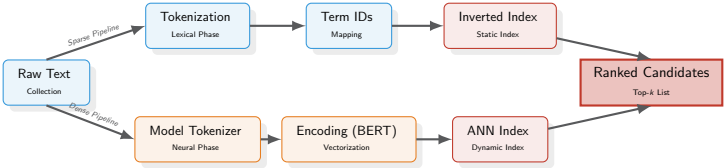

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The Standard ANN Approach: IVF + PQ

1. **Clustering-Based (IVF)**
   - Cluster vectors with k-means
   - Search only in nearest clusters
   - ✔ Fast, scalable
   - ✘ Lower accuracy without tuning
2. **Product Quantization (PQ)**
   - Compress vectors aggressively ($32\times$)
   - Approximate distances in compressed space
   - ✔ Very memory-efficient
   - ✘ Moderate accuracy loss
3. **Combined (IVF + PQ)**
   - Use clustering to narrow search area
   - Use compression to fit index in RAM and speed up distance calculation
   - ✔ **The industry standard for large-scale RAG**

IVF: Inverted File Index (for Vectors!)

Core Idea

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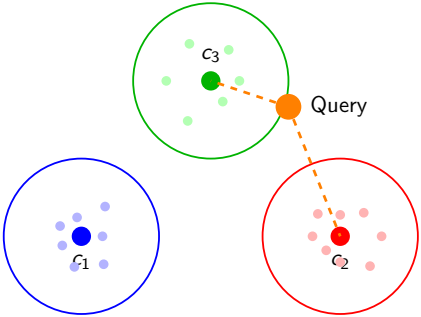

Search only in 1-2 nearest clusters (not all!)

IVF: Construction and Search

Construction

1: Run k-means on all vectors $\rightarrow$ get k cluster centers (e.g., $k = 1000$)
2: Assign each vector to its nearest cluster center
3: Build inverted lists: cluster $i \rightarrow$ [vector IDs in cluster i]

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Search

Input: Query $\vec{q}$, n_{probe} (clusters to search), k
Compute distance from $\vec{q}$ to all cluster centers
Select n_{probe} nearest cluster centers
candidates $\leftarrow$ []
for each selected cluster **do**
    Fetch all vectors in cluster
    Compute exact distances to $\vec{q}$
    Add to candidates
end for
Sort candidates by distance, return top-k

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Input: Query q⃗, n_probe (clusters to search), k
Compute distance from q⃗ to all cluster centers
Select n_probe nearest cluster centers
candidates ← []
for each selected cluster do
    Fetch all vectors in cluster
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end for
Sort candidates by distance, return top-k
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Key parameter: n_{probe} (1 = fast/low recall, 100 = slow/high recall)

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Method:

1. **Split** each vector into m subvectors (e.g., $m = 96$ subvectors of 8 dims)
2. **Quantize** each subvector: assign to nearest centroid (256 centroids)
3. **Store** centroid ID (1 byte) instead of full subvector

Example

Original: [0.23, -0.41, 0.87, 0.12, -0.34, 0.56, 0.78, -0.23, ...] (768 floats)

Split into 96 subvectors:

- Subvec 1: [0.23, -0.41, 0.87, 0.12, -0.34, 0.56, 0.78, -0.23] → Centroid 42
- Subvec 2: [0.45, 0.67, -0.12, 0.89, 0.34, -0.56, 0.23, 0.11] → Centroid 123
- ... (96 total)

Compressed: [42, 123, 7, 89, ...] (96 bytes)

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Can approximate distances using only centroid IDs (no need to decompress!)

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Algorithm:

1. **Precompute** distance table: query subvector to all centroids
   - For each of 96 subspaces
   - For each of 256 centroids
   - Total: $96 \times 256 = 24{,}576$ distances (computed once per query)
2. **For each document:**
   - Approximate distance $= \sum_{i=1}^{96} \text{distTable}_i[\text{code}_i]$
   - Just 96 lookups + additions (very fast!)

Performance

- **Memory:** 32× compression (3GB → 96MB for 1M vectors)
- **Speed:** 10-100× faster than exact distance
- **Accuracy:** 80-95% recall typical

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Architecture:

1. **Offline:** Cluster vectors into k clusters (IVF)
2. **Offline:** Compress vectors with PQ
3. **Online:** Find n_{probe} nearest clusters
4. **Online:** Search within those clusters using PQ distances

Real-World Impact

Facebook/Meta AI Similarity Search (FAISS):

- 1 billion vectors $\times$ 768 dims $\times$ 4 bytes = **3 TB**
- With IVFPQ: 1 billion vectors $\times$ 64 bytes = **64 GB**
- **50$\times$ memory reduction!**
- Still achieves 90-95% recall with proper tuning

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Used by: Meta, OpenAI, many large-scale systems

FAISS Demo: Dense Indexing (IVF+PQ)

```python
import numpy as np
import faiss

# Simulate document embeddings (100k docs, 768 dims)
dimension = 768
n_docs = 100000
doc_embeddings = np.random.randn(n_docs, dimension).astype('float32')
faiss.normalize_L2(doc_embeddings)  # For cosine similarity

# === 1. Training & Building IVF+PQ index ===
n_clusters = 100  # Number of clusters (centroids)
m_subvectors = 96 # Number of PQ subvectors (768/96 = 8 dims each)
bits = 8          # Bits per subvector (256 centroids per subspace)

quantizer = faiss.IndexFlatL2(dimension)
index = faiss.IndexIVFPQ(quantizer, dimension, n_clusters, m_subvectors, bits)

print("Training index...")
index.train(doc_embeddings) # PQ needs training to find centroids
index.add(doc_embeddings)
print(f"Total docs indexed: {index.ntotal}")

# === 2. Search ===
query = np.random.randn(1, dimension).astype('float32')
faiss.normalize_L2(query)

index.nprobe = 10  # Search 10 nearest clusters
distances, indices = index.search(query, k=10)

print("Top 10 document indices:", indices[0])
print("Approximate distances:", distances[0])

# Memory usage (approximation)
# Uncompressed: 100,000 * 768 * 4 bytes = 307.2 MB
# IVFPQ: 100,000 * 96 bytes = 9.6 MB (~32x smaller!)
```

The Decision Matrix: Scale vs. Precision

Aspect	Sparse (Inverted Index)	Dense (IVF+PQ)
Best for	Exact Match, Rare Keywords	Semantic Similarity
Recall	Medium (High for entities)	High (Semantic)
Precision	Very High	High
Latency	5-20ms	10-50ms
Memory	Low	Low-Medium (due to compression)
Scalability	Excellent	Excellent (with IVF)
Explainability	High	Low (Black box)

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- Use **Sparse** when you need to find exact names, model numbers, or unique terms.
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The Challenge:

- Cannot hold 100M raw vectors in RAM (300GB+).
- Users need both exact titles (Sparse) and broad conceptual search (Dense).

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- **Sparse (BM25):** Index title, authors, abstracts for keyword matching.
- **Dense (IVF+PQ):**
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  - Compress vectors 32× (fits on a single server!).

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Result: Billion-scale search with low latency and low cost.

Common Pitfalls and Solutions

Pitfall 1: Using Only Dense for Entity-Heavy Domains

Problem: Query "Python 3.11 release date" $\rightarrow$ matches "snake release from zoo"
Solution: Use hybrid! BM25 catches "Python 3.11" exactly

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Production Checklist

Before deploying, verify:

Sparse Index:

- ☐ Compression enabled
- ☐ Stop words handled
- ☐ Stemming appropriate
- ☐ Field boosting configured
- ☐ Metadata filterable

Dense Index (IVF+PQ):

- ☐ Vectors normalized
- ☐ Number of probes (n_probe) tuned
- ☐ Memory budget met
- ☐ PQ training data is representative

Monitoring:

- ☐ Track recall@k (vs. exact search)
- ☐ Monitor p95/p99 latency
- ☐ Alert on index growth
- ☐ Log query/document drift

Modern IR Infrastructure Examples

Production RAG (Cloud/Managed)

- **Embedding:** OpenAI `text-embedding-3-small`
- **Vector DB:** Pinecone, Qdrant, or Milvus Managed.
- **Candidate Generation:** Dense-first (IVF or HNSW hidden by provider).
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What We Covered Today

1. **Sparse Indexing (Classical)**
   - Inverted index structure
   - Text preprocessing pipeline
   - Compression (Gap encoding + VB)
   - PyTerrier demo
2. **Dense Indexing (Neural)**
   - Why ANN search is needed
   - IVF+PQ: The standard for scale (Clustering + Compression)
   - FAISS demo
3. **Design Decisions**
   - Modern IR Pipeline (Candidate Generation)
   - Traditional IR vs. RAG
   - Scaling to billions of documents

Key Takeaways

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Connection to Next Lectures

Lecture 3: Ranking Models

- BM25 scoring uses inverted indexes we built today
- Query expansion (RM3) operates on sparse retrieval
- Language models for better ranking

Lecture 4: Transformer Re-rankers

- Re-rank candidates from hybrid retrieval
- Cross-encoders (BERT) for precision
- Two-stage: retrieve (today) $\rightarrow$ re-rank (lecture 4)

Lecture 5: Dense Retrieval (Deep Dive)

- Bi-encoders in detail
- Training dense retrievers
- We already know ANN indexing!

Lecture 6: Learning to Rank

- Combine sparse + dense scores optimally
- Learn fusion weights from data
- Put it all together

Assignment 2: From Sparse to Dense Retrieval

Due: 2 weeks

Tasks:

1. **Sparse Indexing (40 pts)**
   - Build inverted index with PyTerrier.
   - Measure effect of compression (VB) on index size.
2. **Dense Indexing (40 pts)**
   - Build IVF+PQ index with FAISS on 100k vectors.
   - Measure recall vs. exact search as you change `n_probe`.
3. **Analysis (20 pts)**
   - Compare latency and memory usage of Sparse vs. Dense.
   - Identify query types where each method fails.

Resources

Documentation

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Papers (Optional Reading)

- HNSW: Malkov & Yashunin (2018) - "Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs"
- Product Quantization: Jégou et al. (2011) - "Product quantization for nearest neighbor search"
- Hybrid Search: Lin et al. (2021) - "A Few Brief Notes on DeepImpact, COIL, and a Conceptual Framework for Information Retrieval Techniques"

Resources

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Questions?

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Office Hours: [Your time]

Email: [Your email]

Next Class: Lecture 3 — Ranking Models

We'll use the indexes we built today to implement BM25, language models, and query expansion!

Start the assignment early — building a hybrid system takes time!